Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **C++/Python systems**, distributed workflows, and performance-sensitive backend services on Linux-oriented stacks.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services** for 500+ users, translating stakeholder requests into APIs, validation logic, and schema updates while supporting structured data workflows, operational tooling, and maintainable production releases across shared systems.
- Improved **system performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and reliability across recurring production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence logic, and automated checks to improve backend reliability, validation coverage, and operational responsiveness across production workflows.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading features, documenting fixes and reducing feedback turnaround by **60%** while improving issue isolation, iteration speed, and maintainability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, data movement, ETL workflows, and reliable release coordination.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions while strengthening defect isolation, reproducibility, and documentation across build and integration cycles.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python pipelines** across 100K+ trajectory logs, improving long-horizon planning by **18%** in simulation while strengthening testing discipline, algorithm validation, and reproducible engineering analysis for autonomous-driving workflows.
- Architected a **spatial-temporal encoder** combining structured map geometry and agent context through multi-head attention, improving closed-loop planning stability by **15%** across repeatable validation, debugging, and simulation-analysis workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows while strengthening high-volume reliability, service coordination, and software-component interoperability.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and engineering handoffs during iterative software delivery.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash
Systems & Infra: Linux, Unix, operating systems, distributed systems, networking, data structures, algorithms, threading models, REST APIs, WebSockets, Docker, Jenkins, CI/CD, Git/GitHub
Engineering: SQL/NoSQL, PostgreSQL, MongoDB, debugging, test automation, simulation workflows, performance tuning, schema design, object-oriented programming, code review, Agile

Projects

CPU Scheduling Simulator Live | *C++, React, JavaScript, OS Algorithms*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, debugging workflows, validation, timing analysis, and interactive visualization across simulated workloads, strengthening systems reasoning and correctness-focused testing practices.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management from a unified interface while working across Java backend and TypeScript frontend modules.

FinNexus Live | *Next.js, WebSockets, Kubernetes, MongoDB*
- Built a **secure financial web platform** supporting 50K+ users and 1M+ transactions, integrating real-time updates, tokenized access controls, and CI/CD workflows that improved system coordination and application reliability.